

Further Human–Computer Interaction

The 80/20 Revision Guide

Part IB — Cambridge CST — Paper 7

How this guide is organised

Further HCI sets two questions a year on **Paper 7**. Across **eight years (2018–2025, 16 questions)**, the tagged frequencies are strikingly concentrated:

Topic	Appearances	Share
Cognitive Dimensions of Notations framework	10	62%
Visual representation theory	5	31%
Design & evaluation methods	5	31%
Bayesian / smart interaction	3	19%
Empirical methods (Fitts, KLM, stats)	3	19%
Graphical user interface design	2	13%
Augmented & virtual reality	2	13%
Generative-AI interfaces	2	13%
Attention Investment model	2	13%
Probabilistic interfaces / HCI waves / voice / IDEs	1 each	6%

80/20 verdict. The **Cognitive Dimensions framework** appears in nearly two-thirds of all questions — usually “apply the CDs to evaluate notation *X*.” If you revise one thing, revise that. **Visual representation** and **evaluation methods** are the next reliable pair (~30% each). Critically, the trendy-looking topics — AR/VR, generative-AI interfaces, voice — are *not* separate theory: the examiner expects you to analyse them **by analogy to the CDs, visual representation, and user/goal/task models**. So the three core frameworks cover ~90% of the paper, the fashionable questions included.

This document is ordered that way: **the three frameworks first, then the quantitative methods, then context**.

1 The three frameworks (90% of marks)

1.1 Cognitive Dimensions of Notations (the headline — 62%)

Exam-critical

The CDs (Green & Blackwell) are a **broad-brush vocabulary for discussing the usability of a notation together with its environment** — usability is a function of *both*. They are built for **interaction spaces** (programming languages, APIs, spreadsheets, CAD) where task-based methods (Cognitive Walkthrough, KLM) suffer “**death by detail**.” Core premise: **there is no perfect notation** — every design is a set of **trade-offs**, and improving one dimension typically degrades another. CDs give designers a shared language to *discuss* those trade-offs, not a score.

The dimensions (be able to define and apply these)

- **Viscosity** — resistance to change (small edit needs many actions).
- **Hidden dependencies** — important links between entities aren't visible.
- **Premature commitment** — forced to make decisions before the needed information is available.
- **Secondary notation** — meaning carried outside formal syntax (comments, layout, colour).
- **Visibility & juxtaposability** — can you see / compare the parts you need at once?
- **Closeness of mapping** — how close the notation is to the domain it describes.
- **Consistency** — similar semantics expressed in similar forms.
- **Diffuseness** — verbosity; how much space an idea takes.
- **Error-proneness** — the notation invites mistakes.
- **Hard mental operations** — high cognitive load to work things out.
- **Progressive evaluation** — can you check incomplete work at any time?
- **Provisionality** — can you sketch / be vague while exploring?
- **Role-expressiveness** — is it easy to infer what each part is *for*?
- **Abstraction** — the abstraction mechanisms available (and whether you're forced to use them).

Notational activities & the CDs “profile.” Which dimensions matter depends on what the user is doing: *search*, *incrementation* (add content, structure unchanged), *modification*, *transcription*, *exploratory design* (incrementation + modification, end-state unknown), *collaboration*. A CDs **profile** emphasises the dimensions relevant to the dominant activity — e.g. exploratory design cares about *viscosity* and *provisionality*; transcription cares about *closeness of mapping*; search cares about *visibility* and *hidden dependencies*.

Gotcha

CDs are a **discussion vocabulary, not a metric** — you don't “score” a notation. Don't recite all 14 mechanically: **pick the 4–6 dimensions relevant to the user's activity**, give a concrete example of each in the named notation, state the **trade-off**, and propose an **environment fix** (e.g. a refactoring tool reduces viscosity; folding improves visibility). That structure is what scores.

1.2 Visual representation (31%)

Exam-critical

Designing a visual display means choosing the **correspondence** between an *invisible information structure* (elements + relationships) and the visible **marks** on a surface. “All visual representations are marks on a surface intended to correspond to things understood by the reader.”

Worth knowing

The unified theory (use it to analyse any display). The two surface dimensions can correspond to: *physical space* (a map), *object dimensions*, a *pictorial perspective*, or *abstract continuous scales* (time/quantity → graphs). The surface is partitioned into **regions** interpreted differently; within a region, elements are **aligned, grouped, connected or contained**

to express relationships. Every correspondence must be understood **by convention or by explanation**.

The catalogue of conventions (cite these by name):

- **Typography & grids** — screen text inherits centuries of paper convention; arrange within a grid.
- **Maps & graphs** — *quantitative correspondence* of a surface direction to a continuous quantity (Playfair).
- **Schematic drawings** — relative scale, *annotations*, *white space* marking interpretive breaks.
- **Pictures & the *resemblance fallacy*** — images rely on shared interpretive *conventions*, not on simulating real perception; “a good picture need not simulate visual experience.”
- **Node-and-link diagrams** — show *connectivity*; node position is free, so it can carry **secondary notation** (London Underground map).
- **Icons & symbols** — semiotics: pictorial / arbitrary / *metonymic* (key = locking) / *mnemonic*.
- **Visual metaphor** (the desktop) — and its **critique**: theories of visual *representation* explain the desktop better than “metaphor”; strict analogy to physical objects becomes obstructive (General Magic, MS Bob failed).

1.3 Design & evaluation methods (31%)

Design is **iterative**: *divergent* phases (invention — theory suggests new options) and *convergent* phases (critique — theory predicts what will work).

Exam-critical

Place every evaluation method on three axes: **Formative** (early, improve the design, open questions) vs **Summative** (late, test against criteria, closed questions); **Analytic** (apply a theory) vs **Empirical** (observe users); **Quantitative** vs **Qualitative**.

	Quantitative	Qualitative
Analytic	KLM / GOMS	Cognitive Walkthrough, Cognitive Dimensions
Empirical	A/B tests, controlled trials (RCT)	think-aloud, interviews, field observation

Worth knowing

Cognitive Walkthrough (analytic, formative): for each step ask — is the user’s *goal* clear, are the necessary *actions available*, are they *labelled* recognisably, does the system give *feedback* of progress?

Controlled experiments & stats: state a **null hypothesis** (no difference); a **significance test** (e.g. *t-test*) checks whether the difference in means is beyond chance; report **effect size** and **confidence intervals**, not just significance. Use a non-parametric **sign test** for non-normal, *within-subject* matched samples. An **RCT** needs a performance measure, a representative sample, and an experimental task; assess **internal** vs **external validity**.

Analysing qualitative data: *categorical coding* (a coding frame; **inter-rater reliability** via Cohen’s / Fleiss’ *Kappa*, 0.6–0.8 = “substantial”); *grounded theory* (open coding → memos → axial coding → constant comparison → **saturation**).

Gotcha

Know the **bad** techniques and name them: the **Hawthorne effect** (performance improves just because users are being studied), **experimental demand** (users try to please you), and **HiPPO** / introspective single-subject opinion. “I find it more intuitive” is not evidence. Match the method to the question: *open* question $\Rightarrow$ qualitative/formative; *closed* $\Rightarrow$ quantitative/summative.

2 Quantitative interaction (Tier 2)

2.1 Designing smart systems / Bayesian interaction (19%)

Smart interfaces (predictive text, recommenders, code completion) build a **probabilistic model of the user’s intention**. Information gain of one event:

$$h(x) = \log_2 \frac{1}{p(x)} = -\log_2 p(x).$$

A uniform 27-key keyboard transfers ≈ 4.8 bits/keypress, but English compresses to ≈ 2.1 bits/char — the gap measures the inefficiency of treating all keys as equally likely (Fitts’s Law then says make frequent keys bigger).

Worth knowing

Bayesian text entry. Rank the next token by its posterior given context, $p(w_i \mid w_1 \dots w_{i-1})$; via Bayes this is $\propto p(\text{evidence} \mid w_i) p(w_i)$, where the **language model is the prior**. Because only *ratios* are needed, the evidence normaliser cancels. **Adaptive UIs** go further — dynamically relayout to maximise information gain — but changing layouts costs relearning (a **power law of practice** trade-off). LLMs are sophisticated priors: they help a user only insofar as the user’s goals match the prior the model encodes.

2.2 Efficient systems: Fitts, KLM (19%)

Exam-critical

Fitts’s Law — time to point at a target of width W at distance D :

$$T = k \log_2 \left(\frac{2D}{W} \right),$$

the log term being the **index of difficulty**. Interpreted information-theoretically: the speed–accuracy trade-off is an information channel (selecting a small target from a wide range = more information). *Semantic pointing* warps the motor–pixel mapping to enlarge likely targets.

Keystroke Level Model (KLM) predicts *expert* task time as a sum of unit operators: keystrokes, pointing (via Fitts), homing hand to mouse/keyboard (≈ 400 ms), and **mental preparation** (≈ 1350 ms); it ignores learning and error. **GOMS** (Goals/Operators/Methods/Selection) refines the mental-preparation estimate.

2.3 Attention Investment & emerging interaction spaces (13% each)

Worth knowing

Attention Investment (a model of abstraction use / end-user programming): automating a task means writing an **abstract specification** (e.g. a regex). The user weighs **cost** (attention spent programming) + **risk** (bugs → more manual fix-up) against **payoff** (future attention saved). This is why non-programmers rationally prefer repeated manual action — *bounded rationality* applies.

Emerging topics, answered through the core frameworks. *AR/VR* is still a **visual representation** (marks/correspondences — and arguably escapes the desktop *metaphor* problem). *Generative-AI / conversational agents*: ask whether they build a **user / goal / task model**, and — crucially — whether the user can *see, test, or modify* it. *IoT / ubicomp*: where is the **notation** for configuring future action? Analyse all three with CDs + visual representation + attention investment.

3 Context (Tier 3 — a paragraph each)

Three waves of HCI. *1st (1980s)*: cognitive science / human factors; goal = *efficiency* on well-defined tasks. *2nd (1990s)*: social science — *ethnography*, contextual inquiry, participatory design; systems as socio-technical. *3rd (2000s)*: culture & creativity — *user experience, discretionary use, speculative design*; evaluate via meaning, not just efficiency. Be able to place a piece of research in its wave.

Goal-oriented interaction & decision models. Model interaction as **search** toward a goal; but users show **bounded rationality** / *satisficing*, modelled by **prospect theory** (utility weighted by likelihood) and **heuristics & biases** (availability, representativeness, loss aversion, bandwagon). **Persuasive design**/nudge tries to change goals (risk: *paternalism*). **Wicked problems** (Rittel & Webber): no definitive formulation, no stopping rule, solutions good/bad not true/false, every attempt counts.

Designing meaningful systems. *Design ethnography* — holistic, in-context study of activities *and* artefacts; “thick” description; **mixed methods** combine qualitative depth with quantitative breadth.

Exam checklist

Exam-critical

1. **Cognitive Dimensions**: define the framework (notation + environment; no perfect notation; trade-offs); know the dimensions; map them to the user’s *activity profile*; remember *secondary notation*. To answer “evaluate notation *X*”: pick 4–6 relevant dimensions, give concrete examples, name the trade-offs, propose an environment fix. *Discuss, don’t score.*
2. **Visual representation**: correspondence between an invisible information structure and marks on a surface; the unified theory (2 dims → space/object/perspective/abstract scale; regions; align/group/connect/contain; by convention or explanation); the *resemblance fallacy*; critique a real display.
3. **Evaluation**: the formative/summative × analytic/empirical × quant/qual cube; Cognitive Walkthrough steps; null hypothesis / t-test / effect size / validity; coding + Kappa + grounded theory; name the Hawthorne effect, experimental demand, HiPPO.
4. **Smart / Bayesian**: $h(x) = \log_2 \frac{1}{p(x)}$; rank next token by $p(w \mid \text{context}) \propto p(\text{ev} \mid w)p(w)$ with the language model as prior; adaptive-UI trade-off.

5. **Efficient:** Fitts $T = k \log_2(2D/W)$ (index of difficulty); KLM operators (expert, no error/learning); GOMS.
6. **Attention Investment** = cost + risk vs payoff of abstraction. **Emerging topics** (AR/VR, GenAI, voice): answer via visual representation + CDs + user/goal/task models — not as separate theory.